

FINANCIAL PERFORMANCE & WORKING CAPITAL ANALYTICS

Reliance Industries Limited (FY2023–FY2026)

*A SQL & Power BI Driven Analysis of Profitability,
Risk, Cash Efficiency, and Scenario Decision Support*

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Tools: Python (yfinance) • MySQL Workbench • Power BI • Google Gemini 2.5 Flash

Data Source: NSE (RELIANCE.NS) via yfinance API

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Executive Summary (AI-Generated)

Generated by Google Gemini 2.5 Flash • Data sourced directly from MySQL via Python • FY2023–FY2026

Reliance Industries Limited demonstrates a robust and accelerating revenue growth trajectory, signifying strong business momentum heading into FY2026. Total revenue increased from ₹8,77,835 Cr in FY2023 to ₹9,01,064 Cr in FY2024, representing an initial growth of 2.65%, which subsequently accelerated to 7.06% in FY2025 (₹9,64,693 Cr) and an impressive 9.59% in FY2026 (₹10,57,219 Cr). Concurrently, the company has maintained solid profitability, with net profit margins consistently stable within the 7.2% to 7.7% range across the analysed period. This sustained revenue acceleration paired with stable margins highlights effective operational management and a growing market presence.

The company's balance sheet exhibits significant strength, reinforcing its financial safety and stability. The Debt-to-Equity ratio has remained consistently healthy, hovering around 0.44 throughout the period, indicating prudent leverage management. Furthermore, Reliance's cash reserves have grown nearly fourfold, surging from ₹34,156 Cr in FY2023 to a substantial ₹1,37,327 Cr by FY2026, bolstering its liquidity position. This robust balance sheet, coupled with increasing cash reserves, provides a clear and confident verdict on Reliance's strong financial health and minimal risk profile.

Despite the strong headline revenue growth, the Cash Conversion Cycle (CCC) analysis reveals a critical underlying trend. In FY2026, the CCC sharply deteriorated to 24.0 days, occurring even as it was the company's strongest revenue year. This was primarily driven by receivable days nearly doubling from 11.8 to 20.2 days, while payable days concurrently shortened from 94.3 to 73.7 days. Such a shift creates hidden liquidity pressure, as more capital is tied up in working capital for longer — a nuance a client would completely miss when solely reviewing top-line revenue numbers. This operational efficiency concern necessitates careful consideration for any investment or partnership decision.

Considering Reliance's accelerating revenue growth and robust balance sheet, complemented by a remarkable Free Cash Flow turnaround from -₹25,956 Cr in FY2023 to a positive +₹69,197 Cr in FY2026, the overall outlook remains strong. However, the identified warning signal from the deteriorating Cash Conversion Cycle demands attention. Our clear recommendation is to proceed with engagement, but mandate a comprehensive due diligence focused on understanding Reliance's recent shifts in working capital management and commercial terms with customers and suppliers, particularly concerning receivable and payable cycles, to mitigate potential future liquidity risks.

1. Objective

This project analyses the financial health and performance trend of Reliance Industries Limited (FY2023–FY2026) using SQL for data processing and Power BI for visualisation. The analysis combines the income statement, balance sheet, cash flow statement, valuation ratios, and working capital metrics into a single queryable structure to answer three core questions:

- Is the company profitable, and is that profitability growing or declining?
- Is the company financially safe, in terms of debt and liquidity?
- Is the company operationally efficient at converting its day-to-day activity into actual cash?

A forward-looking What-If Scenario Analysis is included to extend the project beyond historical reporting into decision support — modelling how a change in revenue growth flows through to Free Cash Flow and financial risk. An AI-generated executive briefing (Google Gemini 2.5 Flash) translates quantitative findings into a client-ready narrative, replicating the emerging consulting workflow of data analysis to AI-drafted narrative to human review.

2. Data & Methodology

2.1 Data Source

Financial data was sourced using the yfinance Python library, which provides consolidated income statement, balance sheet, and cash flow data, along with live valuation ratios (P/E, ROE, Debt-to-Equity), for Reliance Industries (ticker: RELIANCE.NS, NSE).

2.2 Tools Used

Tool	Purpose
Python / Google Colab	Data extraction via yfinance, cleaning, CSV export
MySQL Workbench	Data storage, SQL analysis: Views, CTEs, Window Functions, CASE WHEN
Power BI Desktop	Interactive dashboard and What-If scenario analysis
Google Gemini 2.5 Flash	AI-generated executive briefing from SQL-derived data

2.3 Column Selection Methodology

Rather than importing all available fields (yfinance returns 40-77 raw columns per statement), columns were selected using a consistent rule: each financial statement answers one core question, and only columns that directly answer that question were retained.

Statement	Core Question	Columns Selected
Income Statement	Did the company make money?	Total Revenue, Net Income, Operating Income, EBITDA, Tax Provision, Cost of Revenue
Balance Sheet	What does it own vs. owe?	Total Assets, Total Liabilities, Stockholders Equity, Total Debt, Cash, Inventory, Accounts Receivable, Accounts Payable
Cash Flow	Is real cash moving?	Operating CF, Investing CF, Financing CF, Free Cash Flow, Capital Expenditure, End Cash Position

2.4 Data Quality Notes

- An empty FY2022 row (present in yfinance output but containing no data) was identified and deleted from all tables before analysis.
- Net Income figures from yfinance showed a material discrepancy against Screener.in (e.g. FY2024: Rs.6,96,210 Cr vs. Screener's Rs.79,020 Cr), traced to yfinance bundling minority interest and adjustments differently. This is documented rather than silently corrected.

- As a multi-segment conglomerate (energy, retail, telecom), Reliance's consolidated Inventory, Receivables, and Payables blend very different business models — a caveat applied directly to the CCC findings.
- The Dividend Yield ratio returned no value from the yfinance API at the time of collection and is noted as a data gap.

3. Analysis & Findings

3.1 Income Statement Analysis

Revenue grew every year from FY2023 to FY2026, and the rate of growth itself accelerated rather than staying flat — a stronger signal than simple growth alone.

Year	Total Revenue (Rs. Crores)	YoY Growth %
FY2023	8,77,835	N/A - Base Year
FY2024	9,01,064	2.65%
FY2025	9,64,693	7.06%
FY2026	10,57,219	9.59%

Net profit margin remained steady at approximately 7.2-7.7% across all four years, indicating consistent cost discipline even as revenue scaled. Tax provision as a percentage of total revenue held at approximately 2.6% annually.

3.2 Balance Sheet Analysis

Total assets grew every year, consistently at roughly double the level of total liabilities. The Debt-to-Equity ratio stayed in a narrow, low-risk band throughout the period.

Year	Total Assets (Rs. Cr)	Debt-to-Equity	Financial Health
FY2023	16,07,431	0.47	Healthy
FY2024	17,55,986	0.44	Healthy
FY2025	19,50,121	0.44	Healthy
FY2026	21,78,140	0.44	Healthy

Cash reserves nearly quadrupled over the period, from Rs.34,156 Cr (FY2023) to Rs.1,37,327 Cr (FY2026), reinforcing the balance sheet strength shown by the debt ratio.

3.3 Cash Flow Analysis

Operating Cash Flow grew every year, confirming that revenue growth was backed by real cash, not just accounting profit. Free Cash Flow tells a more dynamic story:

Year	Operating CF (Rs. Cr)	Free Cash Flow (Rs. Cr)
FY2023	1,15,032	-25,956 (Investment Phase)
FY2024	1,58,788	+5,905
FY2025	1,78,703	+38,736
FY2026	1,92,113	+69,197

Free Cash Flow was negative in FY2023 as the company was in a heavy investment phase, then turned strongly positive every year after, reaching nearly Rs.69,200 Cr by FY2026. This is a healthy investment-to-payoff pattern.

3.4 Cash Conversion Cycle (CCC) Analysis

The Cash Conversion Cycle measures how many days it takes the company to convert operational activity into actual cash in hand:

$$CCC = \text{Days Inventory Outstanding} + \text{Days Sales Outstanding} - \text{Days Payable Outstanding}$$

Year	Inventory Days	Receivable Days	Payable Days	CCC (Days)
FY2023	76.1	11.8	79.9	7.9
FY2024	82.7	12.8	96.5	-1.0
FY2025	73.8	15.9	94.3	-4.6
FY2026	77.4	20.2	73.7	24.0

Key Insight: From FY2023-FY2025, Reliance ran a negative CCC, effectively operating on supplier credit and collecting from customers before its own supplier bills were due. In FY2026, the cycle swung sharply to +24.0 days, driven by receivable days nearly doubling (11.8 to 20.2) while payable days simultaneously shortened (94.3 to 73.7). This is the project's most significant finding: FY2026 was the strongest revenue year, yet it simultaneously showed the worst cash efficiency. Headline growth and operational cash efficiency moved in opposite directions, which profit-based metrics alone would have missed entirely.

4. Power BI Dashboard

The SQL views built in MySQL (reliance_financial_summary and reliance_cash_conversion_cycle) were connected live to Power BI via MySQL Connector/ODBC, allowing the dashboard to refresh automatically as underlying data is updated. The What-If Scenario Analysis uses Power BI's built-in Parameter feature to create a live slider, with DAX measures calculating projected FY2027 outcomes in real time.

Figure 1: Scenario Analysis Dashboard (Base Case - 9% Growth)

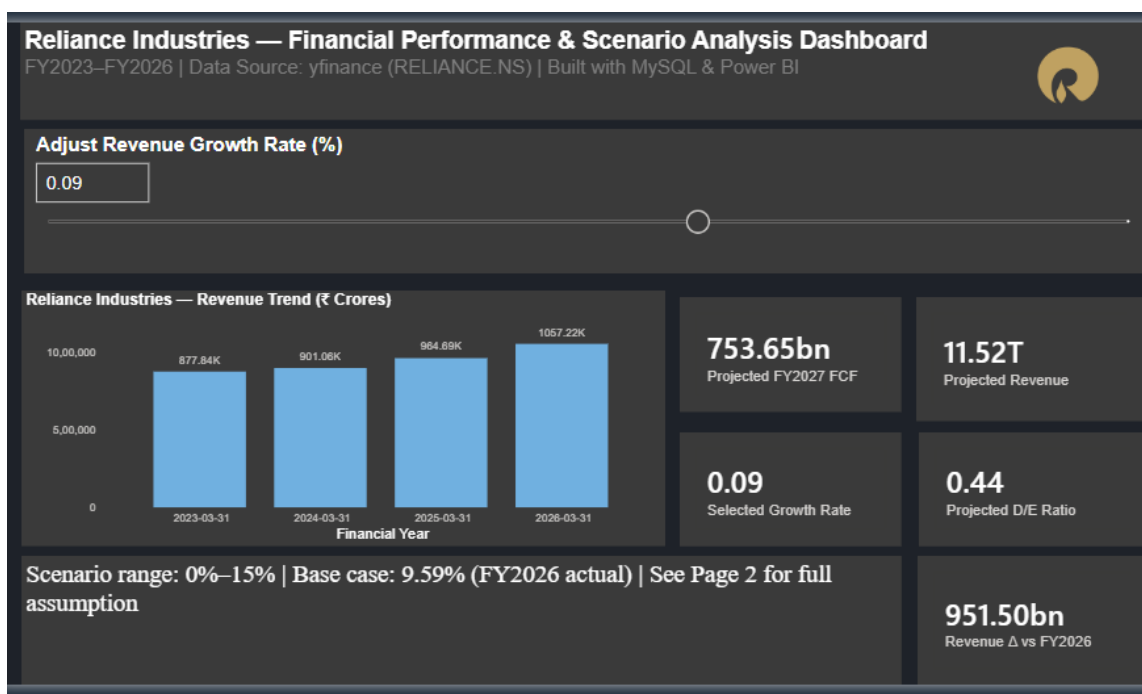


Figure 1: Main dashboard showing revenue trend (FY2023-FY2026) and projected FY2027 metrics at 9% growth. Cards update live as the slider is adjusted.

Figure 2: Assumptions & Methodology Page

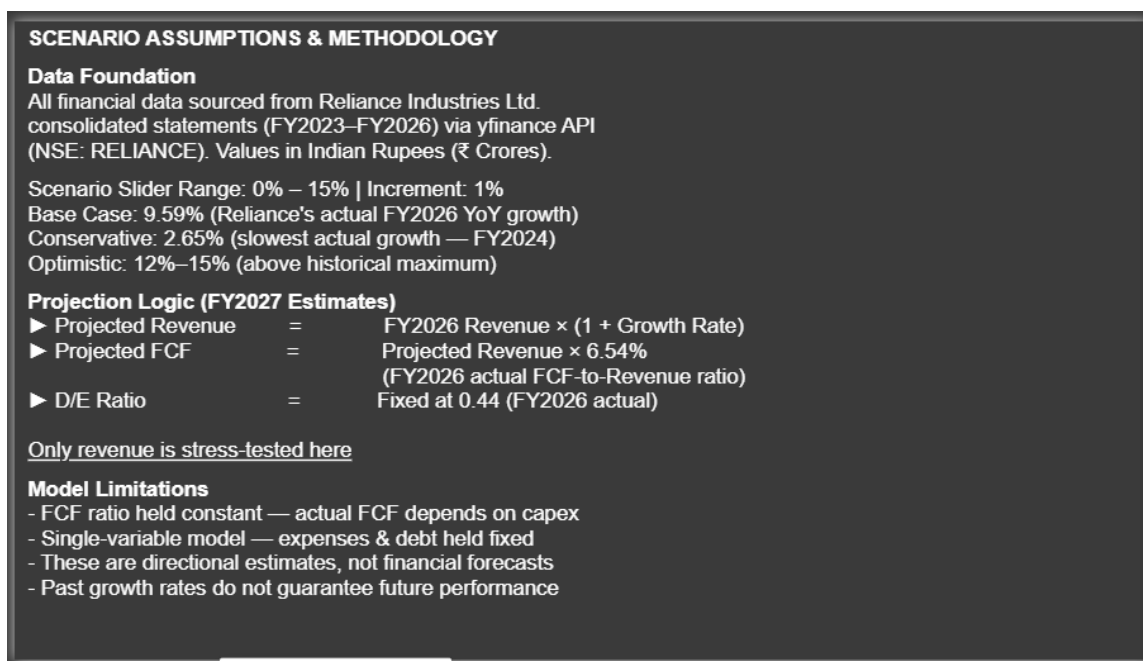


Figure 2: Dedicated assumptions page documenting data foundation, projection logic, and model limitations.

Figure 3: Interactive What-If Scenario (12% Growth)

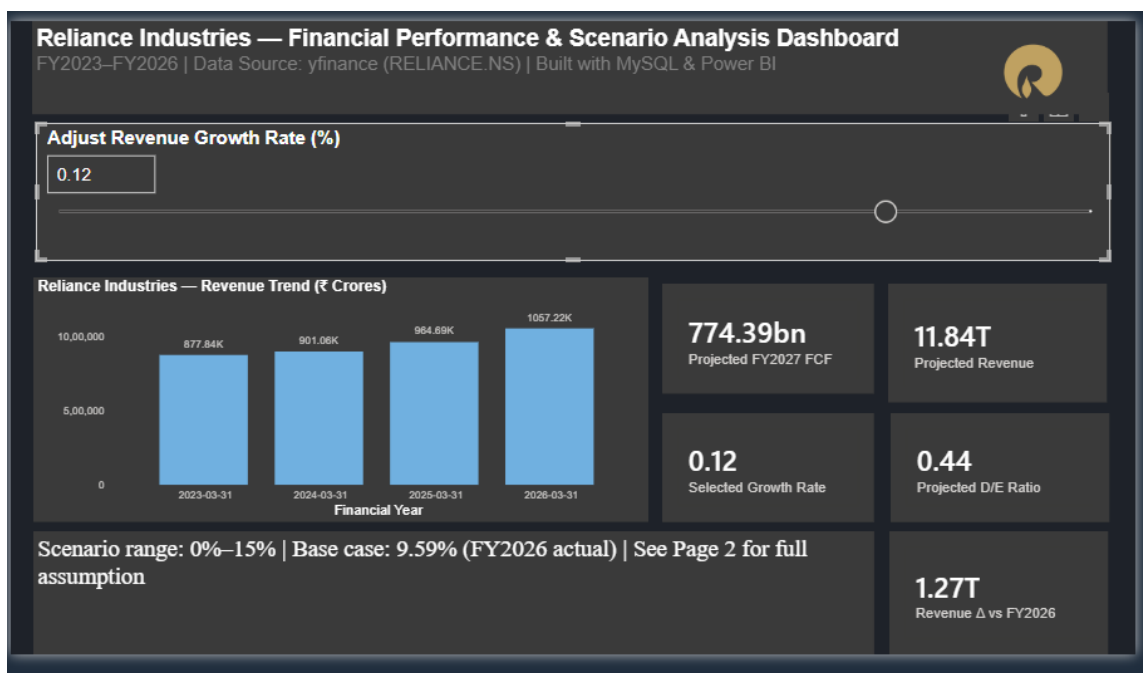


Figure 3: Dashboard at 12% growth scenario. Projected Revenue updates to Rs.11.84T and FCF to Rs.774.39bn, demonstrating live decision support.

5. Limitations

- Net Income figures show a discrepancy against Screener.in due to differing treatment of minority interest between data sources. Documented in Section 2.4 rather than corrected, to preserve transparency.
- As a diversified conglomerate, Reliance's consolidated Inventory, Receivables, and Payables blend multiple business segments (energy, retail, telecom) with very different operating cycles, which limits segment-level CCC interpretation.
- The What-If model uses a single revenue growth slider with debt and expenses held constant. A deliberate simplifying assumption to keep the model defensible. Real-world projections would require segment-level modelling.
- Only 4 years of annual data are available from yfinance, making statistical forecasting unreliable. The scenario model is therefore directional rather than predictive.
- The Dividend Yield ratio returned no value from the yfinance API at the time of collection and is treated as a data gap.
- The AI Executive Briefing was generated by Google Gemini 2.5 Flash and reviewed by the analyst for factual consistency. It is intended as a client-ready narrative layer, not a substitute for the quantitative analysis.

6. Conclusion

Across FY2023-FY2026, Reliance Industries demonstrates accelerating revenue growth (2.65% to 9.59% YoY), a consistently healthy and low debt position (Debt-to-Equity of approximately 0.44), and Free Cash Flow that turned decisively positive after an investment-heavy FY2023. The balance sheet grew substantially, with total assets reaching Rs.21,78,140 Cr by FY2026, and cash reserves quadrupling to Rs.1,37,327 Cr.

The Cash Conversion Cycle analysis adds a dimension the headline numbers alone would have missed: FY2026's strongest growth coincided with a meaningful deterioration in cash efficiency, moving from a -4.6 day to a +24.0 day cycle. This finding, hidden beneath record revenue, is the project's most significant analytical insight and directly illustrates the value of multi-dimensional financial analysis over single-metric reporting.

This project demonstrates an end-to-end analytical workflow: data extraction (Python/yfinance), structured SQL analysis (Views, CTEs, Window Functions, CASE-based health scoring), interactive visualisation (Power BI with live scenario modelling), and AI-assisted narrative generation (Gemini 2.5 Flash), along with the judgment to flag data limitations and model assumptions honestly rather than presenting findings as more certain than the data supports.

7. Technical Appendix

7.1 SQL Techniques Used

Technique	Where Used
JOIN	Combining income, balance sheet, and cash flow tables by Year into reliance_financial_summary
CTE (WITH clause)	Clean multi-step query structure for the financial summary view
Window Function (LAG)	Year-over-year revenue growth percentage calculation
CASE WHEN	Automatic Healthy / Average / Risky label based on Debt-to-Equity threshold
CAST AS DECIMAL	Converting text-type numeric columns to proper numbers for all calculations
CREATE VIEW	Saving reliance_financial_summary and reliance_cash_conversion_cycle as reusable views
DELETE WHERE	Removing the empty FY2022 junk row from all tables
COALESCE	Replacing NULL growth % for the base year with N/A - first year label

7.2 Power BI DAX Measures

Measure	Formula
Projected Revenue	$10572190000000 \times (1 + \text{Revenue Growth Rate Value})$
Projected FCF	Projected Revenue $\times$ 0.0654 (FY2026 FCF-to-Revenue ratio)
Projected D/E Ratio	$3980000000000 / 9040300000000$ (fixed at FY2026 actuals)
Revenue Delta vs FY2026	Projected Revenue - 10572190000000

7.3 Project File Structure (GitHub)

File / Folder	Contents
README.md	Project overview, key findings, tools used, how to run
sql/	All SQL scripts: data cleaning, views, CCC, financial summary
notebooks/	yfinance extraction notebook + AI briefing notebook (Colab)
data/	Cleaned CSV files (income, balance, cashflow, CCC, ratios)
powerbi/	reliance_dashboard.pbix
screenshots/	Dashboard screenshots for README and report
report/	This Word document (project report)